

TEJAS PANDYA

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	Expected 05/27
Master of Science in Financial Engineering GPA: 3.8/4.0	
Graduate Teaching Assistant: Financial Economics; C++ Financial Software Laboratory; Value Investing	
SVKM'S NMIMS, SCHOOL OF BUSINESS MANAGEMENT , Mumbai, India	04/22
Master of Business Administration, Concentration in Finance GPA: 3.3/4.0	
VELLORE INSTITUTE OF TECHNOLOGY , Vellore, India	06/20
Bachelor of Technology in Mechanical Engineering GPA: 8.9/10.0	

PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

Programming: Python, C/C++ (C++20, STL), SQL, Bash, LaTeX/Typst
Libraries & Frameworks: NumPy, Pandas, Polars, SciPy, scikit-learn, statsmodels, LightGBM, XGBoost, PyTorch, pytest
Data & Tools: Linux/Unix, Git, CMake, Streamlit, Shiny, Excel, Bloomberg Terminal, Tableau, Power BI
Certifications: CFA Level I; C++ Programming for Financial Engineering (Baruch College); Bloomberg Market Concepts; CFI FMVA

COURSEWORK HIGHLIGHTS

Quantitative Methods & Computing: Financial Econometrics, Machine Learning in Financial Engineering, Quantitative Methods in Finance, Financial Computing, Applied Numerical Methods, Data Structures and Algorithms
Risk, Valuation & Derivatives: Financial Risk Management, Valuation for Financial Engineering, Quantitative Portfolio Management, Options Pricing & Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives

EXPERIENCE

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	05/26 - 08/26
<i>Quantitative Researcher, QWIM Industry Capstone</i>	
- Lifted Sharpe from 0.37 to 0.55 against a 60/40 benchmark over 1999-2025 as one of five on the QWIM capstone, pairing equal-risk-contribution weighting with hierarchical budgeting on Conditional Drawdown-at-Risk and blending regime-specific covariance by one-month-ahead probability - Implemented a log-periodic power-law singularity bubble-confidence tilt as the fourth ablation level above regime-conditioned covariance blending and quantile-regression tail estimation, the strongest of eight variants, holding Sharpe 0.56 against 0.50 on the 2008-2025 window - Built the Regime Risk Budgeting dashboard tab and its Typst client-report exporter, 1,058 lines of Python across ten modules with twelve unit tests, written fail-safe so that any missing artifact degrades only that one report section rather than breaking the whole client report	
NOMURA SERVICES INDIA PRIVATE LIMITED , Mumbai, India	08/22 - 07/25
<i>Senior Risk Analyst, Market Risk Infrastructure</i>	
- Owned the lifecycle of VaR and sensitivity models across equities, rates, FX and commodities, from daily production runs through backtesting; Python anomaly detection triaged exceptions to stale marks, erroneous data or booking errors, cutting investigation turnaround 40% and restoring SLAs - Delivered quarterly Basel 2.5 and FRTB stress-testing and capital submissions alongside front-office and regulatory teams, engineering rules-based validation over delta, vega and curvature sensitivities that reconciled fully against front-office pricing and caught anomalies pre-submission - Engineered Python and SQL pipelines across equities, rates, FX and commodities that aggregate, validate and reconcile daily P&L, positions and Level-1/Level-2 market data into the VaR and sensitivity dashboards, eliminating more than 50 hours of manual processing every month - Led a three-person workstream on US-region end-of-day risk reporting, cutting overnight batch runtime 35% through SQL query optimisation and job scheduling, and built the VaR, P&L and exception dashboards that senior stakeholders read each morning before the US open	

ACADEMIC PROJECTS

CROSS-SECTIONAL EQUITY ALPHA RESEARCH	07/26
<i>Python, LightGBM, statsmodels</i> github.com/6ixE11even/equity-xs-alpha	
- Constructed eight cross-sectional signals across 495 S&P 500 names, scored by Spearman rank information coefficient with Newey-West t-statistics under Benjamini-Hochberg false-discovery control; only Amihud illiquidity survived at 5.32, while the low-volatility and lottery signals inverted - Scored 15,575 EDGAR filings by Loughran-McDonald tone and traced that factor's 1.37 t-statistic to a 10-K versus 10-Q calendar artefact; differencing within form collapsed it to 0.46, while a LightGBM composite reached 0.79 Sharpe against -0.57 on a survivorship-biased panel	
MACRO-REGIME TACTICAL ASSET ALLOCATION	12/25 - 07/26
<i>Python, cvxpy, MOSEK, scikit-learn</i> github.com/6ixE11even/macro-regime-allocation	
- Detected macro regimes from 120+ FRED-MD indicators by PCA to 95% of variance into two-step K-Means, then benchmarked that detector against Wasserstein K-medoids, a Gaussian HMM and NBER dating across 798 months, finding it switches every 3.8 months against the 44.7 NBER takes - Ran 494 convex solves across 247 out-of-sample months with none failing to reach optimality, charging turnover and quadratic market impact inside the objective rather than subtracting realised costs from returns after the optimiser had already chosen its weights for the month	
ALMGREN-CHRISS OPTIMAL EXECUTION	05/26
<i>Python, NumPy, Deribit API</i> github.com/6ixE11even/optimal-execution	
- Implemented the closed-form Almgren-Chriss liquidation schedule minimising expected cost plus lambda times variance, calibrated to live Deribit BTC-PERPETUAL volatility and average daily volume; it takes about 38% off TWAP's cost standard deviation, not a lower expected cost - Trained a NumPy policy-gradient agent with hand-derived gradients checked against central differences in continuous integration, on 207,361 real five-minute bars; it converged to within 0.060bps of the closed form it was never shown, and nothing beat TWAP out of sample	